Journal of Soil and Water Conservation

ISSN: 0022-4561 (Print) 1941-3300 (Online) Journal homepage: www.tandfonline.com/journals/uswc20

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To cite this article: J.D. Williams, C. Reardon, S.B. Wuest & D.S. Long (2020) Soil water infiltration after oilseed crop introduction into a Pacific Northwest winter wheat–fallow rotation, Journal of Soil and Water Conservation, 75:6, 739-745, DOI: 10.2489/jswc.2020.00165

To link to this article: https://doi.org/10.2489/jswc.2020.00165

Published online: 01 Nov 2020.

RESEARCH SECTION

doi:10.2489/jswc.2020.00165

Soil water infiltration after oilseed crop introduction into a Pacific Northwest winter wheat–fallow rotation

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Abstract: Rapid water infiltration is important for improving water storage and reducing erosion potential in arid and semiarid areas. Taprooted crops are sometimes credited with improving water infiltration. In a study repeated over three years, we tested whether adding a single brassica crop year to the traditional wheat (*Triticum aestivum*)–summer fallow rotation would improve water infiltration. We found significantly faster rates of ponded water infiltration in subsequent wheat crops following canola (*Brassica napus*; 29.68 ± 19.83 mm h^{-1} [1.17 ± 0.78 in hr^{-1}], n = 24) and mustard (*Brassica carinata*; 25.27 ± 17.32 mm h^{-1} [0.99 ± 0.68 in hr^{-1}], n = 19) as compared to wheat following winter wheat (21.14 ± 15.78 mm h^{-1} [0.83 ± 0.62 in hr^{-1}], n = 24). The greater infiltration rates measured following brassicas compared to wheat were not attributed to root channels due to the lack of visible channels. Crop diversification with oilseeds can increase water infiltration after only a single crop year.

Key words: canola—crop diversification—Ethiopian mustard—water infiltration—wheat yield—winter wheat

Wheat (*Triticum aestivum*) production in the intermediate precipitation (300 to 450 mm [11.81 to 17.72 in]) zone of north-central Oregon continues to rely heavily on the two-year winter wheat–summer fallow system to produce stable crop yields. Although economically viable, this system when used with tillage reduces soil aggregation thus impeding water infiltration and soil water recharge leaving the soil susceptible to loss by wind and water erosion (Sharratt and Schillinger 2018; Williams et al. 2018; Williams and Wuest 2011; Williams et al. 2014). In the intermediate to high (450 to 600 mm) annual precipitation zones of the Pacific Northwest, water erosion is the main environmental problem challenging farming and ecology (Shillinger et al. 2010) with snowmelt and frozen soils accounting for greater than 80% of soil loss events (Zuzel et al. 1982). Water infiltration is an important soil process to control water erosion, runoff, and leaching.

Crop intensification to reduce the number of fallow periods and conservation tillage, such as minimum or no-till methods, have been effective in decreasing wind and water erosion (Feng et al. 2011; Williams et al. 2014; Singh et al. 2012), increasing water use efficiency and infiltration, and reducing the decline of soil quality (Franzluebbers 2002; Schillinger et al. 2010). Based on a long-term study in Moro, Oregon (annual precipitation 291 mm [11.46 in]), crop intensification in conjunction with conservation tillage increased soil organic carbon (C) (Machado et al. 2006), which in turn should contribute to aggregate stability and ultimately improve water infiltration and soil nutrient content (Lenssen et al. 2014; Shaver et al. 2002). Anecdotally, producers in the low-to-intermediate precipitation zones of the Pacific Northwest, United States, have noted improved soil structure following brassica oilseed crops (e.g., canola [*Brassica napus*] or Ethiopian mustard [*Brassica carinata*]) with descriptions of soil being easier to drill into for seeding and having better soil tilth and surface characteristics (Sowers et al. 2012). These observations are supported by a study in Australia, where the aggregation of surface soil was improved after only one season of canola production (Chan and Heenan 1991). Improved soil aggregation ultimately improves water infiltration by reducing soil surface sealing by slaked silt particles in loessal soils (Wuest et al. 2005). Increased infiltration following brassicas was demonstrated in an on-farm test in the Pacific Northwest where rates were increased twofold by brassica cultivation as green manure (McGuire 2003).

Plants with taproots, such as brassicas, have the potential to penetrate and loosen compacted soil (Bengough and Mullins 1990; Cresswell and Kirkegaard 1995). This process can be especially useful in no-tillage systems because roots from the previous years have the potential to decay and create long-term biopores (Stirzaker and White 1995; Williams and Weil 2004). Biodrilling involves the extension of roots through compacted soil. The continuous holes left after root decay provide pathways for subsequent roots or water infiltration and percolation. This process has the potential to improve infiltration, tilth properties for other plant roots, nutrient uptake, and soil water storage (Clark et al. 2003). Not all plant roots are equal, however, with size and vigor of a plant playing a role, and some plants more able to penetrate compacted or dense soils depending on limited but preexisting pore size (Bengough and Mullins 1990; Clark et al. 2003). Larger rather than smaller diameter roots more effectively penetrate through dense soil (Materechera et al. 1991; Misra et al. 1986a, 1986b).

Research conclusions on the influence of canola on soil properties are mixed, and it is not clear that it always improves soil structure. Cresswell and Kirkegaard (1995) reported that canola took advantage of existing pores for root development into a compacted B horizon of a red brown earth (Natric Palexeralf) at Temora, New South Wales, but was not responsible for creating substantial new pathways for subsequent

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Received September 12, 2019; Revised January 28, 2020; Accepted March 4, 2020; Published online October 6, 2020.

root development. Chan and Heenan (1991) observed greater soil aggregation following canola compared to barley (*Hordeum vulgare*), which they attributed to the physical characteristics of the plant rooting-patterns and densities rather than microbial drivers (i.e., mycelial networks, glomalin, and bacterial polysaccharides [Aspiras et al. 1971; Bathke and Blake 1984; Caesar-TonThat 2002; Caesar-TonThat et al. 2007; Degens 1997; Oades 1984; Reid and Goss 1981]). However, in an exceptionally dry system with less than 300 mm (11.81 in) precipitation per year, Williams et al. (2018) were unable to discern any differences in soil aggregation between oilseed and cereal rotations.

The objective of this study was to evaluate whether oilseed crops impart a durable effect on soil water infiltration. Ponded water infiltration rates were measured in spring wheat plots that followed either wheat, canola, or mustard. In this study, our hypothesis was that diversification of a winter wheat–fallow rotation with oilseeds (winter wheat–oilseed–fallow) would improve water infiltration in the dryland cropping system of the Pacific Northwest, United States.

Materials and Methods

This research was conducted at the USDA Agricultural Research Service Soil and Water Conservation Research Center, located 15 km (9.3 mi) northeast of Pendleton, Oregon (45°43' N, 118°38' W), at an elevation of 458 m (1,503 ft). This site lies within the intermediate precipitation zone (300 to 450 mm [11.8 to 17.7 in]) of the Pacific Northwest. Although the mean crop year (September 1 through August 31) precipitation from 1930 through 2018 was 413 ± 81 mm (16.26 ± 3.19 in), this site lies within the cultural-practice boundary between annual crop rotations and biennial wheat–fallow production. With a Mediterranean climate, 70% of the precipitation falls between November and April with a minimum of 245 mm (9.65 in) and a maximum of 609 mm (23.98 in) annual totals. The long-term summer precipitation from July through September is 37 mm (1.5 in). Snow cover that might occur anytime from late September through late May is transient due to warm maritime fronts that produce low-intensity storms. Historically, air temperatures have ranged from a minimum (–34.5°C [–30°F]), to maximum (46.1°C [115°F]), with a mean annual air temperature of 10.2°C (50°F). Frost-free conditions range from 135 to 170 days between May and September. A meteorological station located at the site recorded precipitation, wind speed and direction, solar radiation, air and soil temperature, and relative humidity for each crop year of this study. The soil type is a Walla Walla silt loam, (coarse-silty, mixed, mesic, superactive Typic Haploxerolls [United States]; Kastanozems [Food and Agriculture Organization]) containing 21% fine to very fine sand, 69% silt, 10% clay, and positioned on zero slope (Johnson and Makinson 1988). Plots were established where various wheat and broadleaf research has been conducted since 1929.

The treatments were winter wheat, canola (*B. napus*), or spring mustard (*B. carinata*) (table 1). The field trial was designed to determine if the treatment crops influenced the soil properties (i.e., infiltration) and whether effects could be measured with increasing distance from the treatment crops. Plots were arranged in four randomized complete blocks, and the study was repeated for three consecutive years in different areas of a field. Plots were 15 m (49.2 ft) long and

Table 1

Crop seeding sequence and rates to evaluate the effect of oilseed (*Brassica napus* or *Brassica carinata*) incorporation into traditional winter wheat (*Triticum aestivum*) rotation in the 300 to 450 mm y^{-1} precipitation zone of the inland Pacific Northwest United States.

Crop year	Crop	Seeding date	Rate (kg ha^{-1})	Rate (seed m^{-2})
2015	*B. napus* – Amanda winter canola	June 30, 2014	6.7	65
	T. aestivum – Bobtail soft white winter	Oct. 30, 2014	132.3	275
	T. aestivum – Babe soft white wheat spring	Mar. 14, 2015	117.5	312
	B. napus – L130 spring canola*	Mar. 14, 2015	6.2	91
	B. carinata – A110 Ethiopian mustard	Mar. 14, 2015	6.4	91
2016	*T. aestivum* – Ovation soft white winter	Oct. 23, 2015	132.3	254
	B. napus – Amanda winter canola	Oct. 23, 2015	6.7	65
	T. aestivum – Louise soft white spring†	Mar. 17, 2016	134.5	292
	B. carinata – A110 Ethiopian mustard	Mar. 17, 2016	6.4	91
	B. napus – L130 spring canola*	Mar. 17, 2016	6.2	91
2017	*T. aestivum* – Bobtail soft white winter	Sept. 12, 2016	132.3	283
	B. napus – Amanda winter canola	Sept. 21, 2016	6.7	65
	T. aestivum – Louise soft white spring†	Apr. 17, 2017	149.1	308
	B. carinata – 044 Ethiopian mustard	Apr. 17, 2017	6.2	118
2018	*T. aestivum* – Bobtail soft white winter	Oct. 27, 2017	116.6	267
	T. aestivum – Louise soft white spring†	Mar. 27, 2018	129.1	281

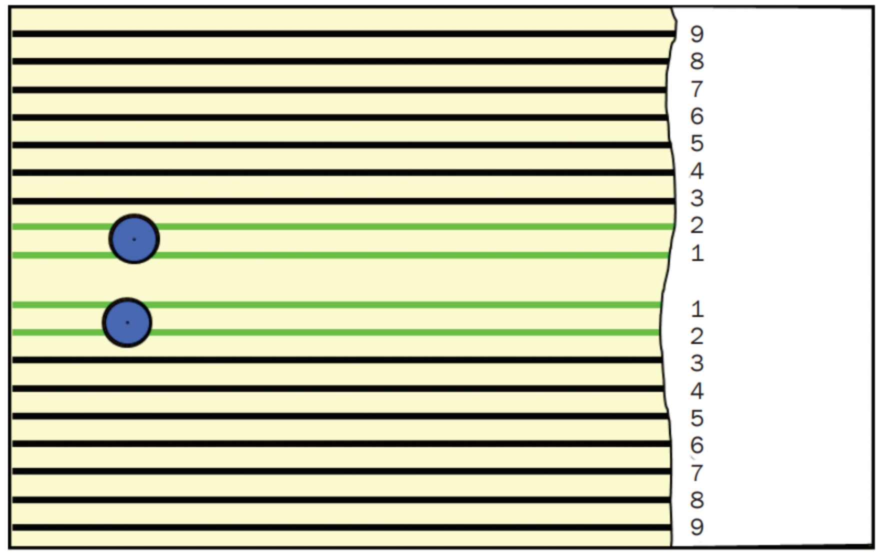
*Reseeding of failed winter canola crop.

†Ring infiltrometer measurements made after harvest.

3.8 m (12.5 ft) wide and seeded using a double disc drill (Great Plains 606 NoTill, Great Plains Ag, Salina, Kansas). The treatment crops winter wheat and winter canola were seeded into plots that had been fallow for 12 to 14 months, whereas spring mustard was seeded into winter wheat plots harvested the prior year. Treatment crops were seeded in four rows surrounded by seven border rows of winter wheat for winter-seeded treatment crops or spring wheat for the spring-seeded treatment crop (figure 1). Winter canola failed to establish in the fall of 2015 and 2016 and was reseeded to spring canola the following spring. All plots were planted to spring wheat eight to nine months (March to April) after harvest to provide a more uniform surface residue and soil moisture condition for infiltration measurements. The spring wheat was seeded into the existing furrows. Spring wheat yields were measured in the center four rows which had been planted to the treatment crops by hand-harvesting two 1 m^2 (10.74 ft^2) bundle samples from each plot.

Figure 1

Treatment furrows and ring infiltrometer placement in oilseed evaluation. Black lines represent furrows always planted to wheat; green lines were furrows planted to canola, mustard, or winter wheat in the treatment phase of the three-year fallow–treatment–spring wheat or wheat–treatment–spring wheat rotation. Blue circles represent placement of ring infiltrometers on furrow pairs following spring wheat harvest. The drawing is not to scale.



Ponded water infiltration rates were measured following spring wheat harvest on top of the treatment furrows using two single-ring infiltrometers per plot. Measurements of spring wheat following canola, mustard, or winter wheat are hereby referred to as canola treatment, mustard treatment, and winter wheat treatment. Sharpened metal cylinders 30 cm (11.81 in) in diameter were driven 25 cm (9.84 in) deep into the soil (Bertrand 1965), always including two crop furrows inside the sample area. Rings were installed by hand in 2016 using a 15 kg (33 lb) slide hammer. In 2017 and 2018 a tractor mounted hydraulic post driver (Shaver H-8) was used for driving the cylinders. The inside circumference was tamped to seal any gaps between the cylinder and the soil. Water was maintained at a constant depth of 2 to 3 cm (0.78 to 1.18 in) with float valves for two hours. Readings from calibrated reservoirs provided periodic estimates of the water infiltration rate (mm h^{-1} [in hr^{-1}]). Two hours has been shown adequate to achieve near steady-state infiltration in these soils (Wuest et al. 2005, 2006). At the end of two hours, a concentrated solution of blue dye (C.I. Food Blue 2, C.I. 42090) was added to bring the water to a concentration of >4 g L^{-1} (>0.53 oz gal^{-1}). After 10 minutes the dyed water was vacuumed from the soil surface, the cylinder pulled from the soil and set on its side, and the underside visually examined to look for preferential flow paths (Wuest 2001b). In rings with exceptionally high infiltration rates, the bottoms of the rings were examined to determine if there were rodent burrows, and as a result, four data points were eliminated.

Data were analyzed using Microsoft Excel descriptive statistics package (Analysis ToolPak, Microsoft Office 365 ProPlus, 2019) and subsequently log transformed to meet assumptions of normality (Gbur et al. 2012). A generalized linear mixed model ("lme4" [R Development Core Team 2012]) was used on log-transformed data to evaluate the effect of the two oilseeds versus winter wheat on infiltration rates after a subsequent spring wheat crop over three years of data collection. Year and block within year were random effects. An orthogonal contrast ("lme4") was used to compare winter wheat against the combined brassica species.

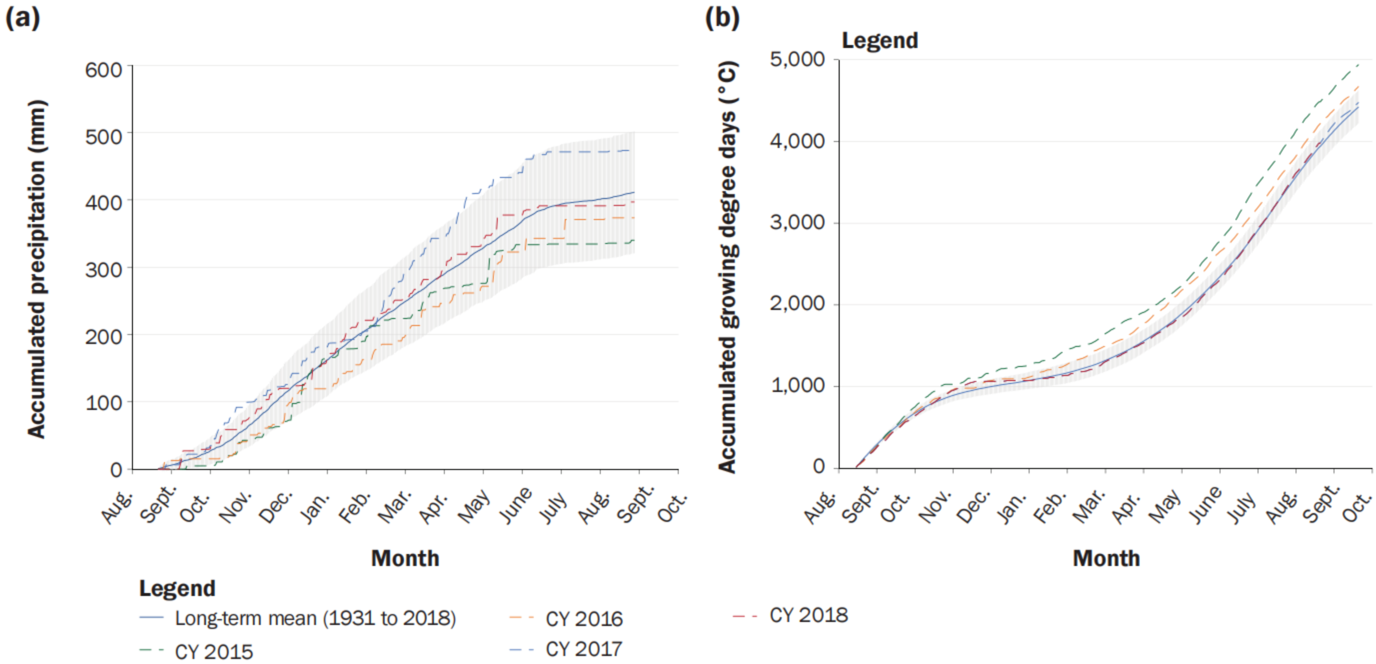
Results and Discussion

Precipitation was within one standard deviation of the long-term mean for each of the treatment crop years (2015, 2016, 2017) and following spring wheat years (2016, 2017, 2018) (figure 2a). However, the accumulated growing degree days exceeded the long-term average for crop year 2016 and even more so in crop year 2015 (figure 2b). Mustard, canola, and winter wheat treatment crop yields were typical for the region (table 2). Spring wheat yields of furrows previously planted to treatment crops mustard, canola, or winter wheat were not significantly different in any given year or when pooled for the three years of research (table 2). Spring wheat yields in 2017 were significantly less ($p \leq 0.05$) than in 2016 or 2018. An analysis (log-transformed) of pooled spring wheat yields from both brassica treatments versus the winter wheat treatment also revealed no difference (oilseed = 4.79 ± 0.23 Mg ha^{-1} [2.14 ± 0.10 tn ac^{-1}], winter wheat = 4.86 ± 0.31 Mg ha^{-1} [2.17 ± 0.14 tn ac^{-1}], Adj. p = 0.89).

Infiltration rates measured in spring wheat stubble of canola, mustard, and winter wheat treatments ranged from 12.04 to 138.49 mm h^{-1} (0.47 to 5.45 in hr^{-1}) with a positive skew to greater infiltration rates (figure 3). Infiltration rates in the canola treatment (29.68 ± 19.83 mm h^{-1} [1.17 ± 0.78 in hr^{-1}], n = 24) were greater (p = 0.02) than in the winter wheat treatment (21.14 ± 15.78 mm h^{-1} [0.83 ± 0.62 in hr^{-1}], n = 24), whereas the mustard treatment infiltration rates (25.27 ± 17.32 mm h^{-1} [0.99 ± 0.68 in hr^{-1}], n = 19) were not significantly different from canola (p = 0.72) or winter wheat (p = 0.18) (figure 3). Water infiltration rates in combined

Figure 2

Two-year accumulations of (a) precipitation and (b) growing degree days (base 0°C). One standard deviation based on records beginning in 1930 are shown around the long-term mean. CY = crop year.



canola and mustard treatments were significantly greater than in the winter wheat treatment (orthogonal contrast, $p < 0.02$). The measured infiltration rates were similar to values recorded in previous research in this region (Williams and Wuest 2011; Wuest 2001b, 2005; Wuest et al. 2006). No significant differences were observed in infiltration rates measured in rings installed manually versus rings installed using the hydraulic post driver ($p = 0.25$).

When we examined the cylinders of soil exposed after infiltration measurements, we did not find evidence of large root material or preferential flow through biopores at the bottom of the cylinders (figure 4). Assessing visually, there was generally more saturated soil at the bottom of the canola and mustard treatments than the winter wheat treatment.

Water infiltration rates were greater in spring wheat following oilseed rather than spring wheat following wheat despite difficulty getting canola to establish. Successful establishment of winter canola in this region depends on placing seed into moist soil at the bottom of deep furrows near the end of a dry summer (Schillinger and Paulitz 2018). The below-average precipitation in 2014 and 2015 produced marginal conditions for establishment of the winter canola crops and resulted in stand failures. Thus, the winter canola treatment was reseeded with spring canola for crop years 2015 and 2016. In the fall of 2016 when the last winter canola crop was planted, precipitation and growing degree days were both well within the long-term normal range, and establishment of the crop was successful.

Reseeding canola in the spring of 2015 and 2016 added an additional disturbance to the soil surface, which would potentially affect the subsequent infiltration results. The

Table 2

Crop yields harvested from treatment furrows for spring wheat or preceding mustard, canola, or winter wheat in the 300 to 450 mm y^{-1} precipitation zone of the inland Pacific Northwest United States. Means and standard errors are shown. There was no statistically significant difference between treatments ($p < 0.05$).

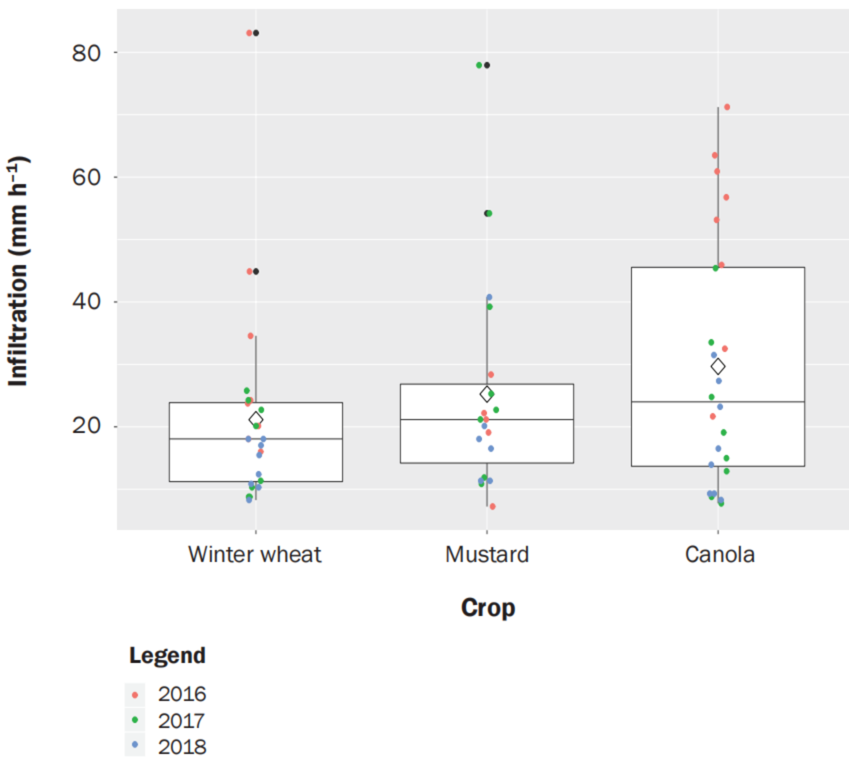
	Crop year (spring wheat)			
	2016	2017	2018	2016 to 2018
Treatment	Yield (Mg ha^{-1})	Yield (Mg ha^{-1})	Yield (Mg ha^{-1})	Yield (Mg ha^{-1})
Mustard*/spring wheat	1.07 ± 0.06/5.64 ± 0.03	3.79 ± 0.04/3.24 ± 0.32	ND†/5.34 ± 0.41	4.56 ± 0.33
Canola/spring wheat	6.14 ± 1.19/5.51 ± 0.15	7.82 ± 0.10/3.22 ± 0.11	ND/6.24 ± 0.36	4.99 ± 0.30
Winter wheat/spring wheat	5.65 ± 0.77/5.94 ± 0.23	8.53 ± 0.98/3.08 ± 0.34	6.92 ± 0.17/5.57 ± 0.36	4.86 ± 0.32

*Mustard, canola, and winter wheat values from the crop year preceding spring wheat production.

†Missing data.

Figure 3

Ponded infiltration rates (total $n = 67$) following harvest of spring wheat planted directly into furrows from which mustard, canola, or winter wheat had been harvested the previous year. Canola versus winter wheat and the contrast between the two brassica species versus winter wheat were significantly different (both at $p < 0.02$). The open diamonds show the mean infiltration rates, and the box plots show the median, 25th, and 75th percentiles. Whiskers indicate furthest data point within 1.5 times the interquartile range, and black dots show outliers.



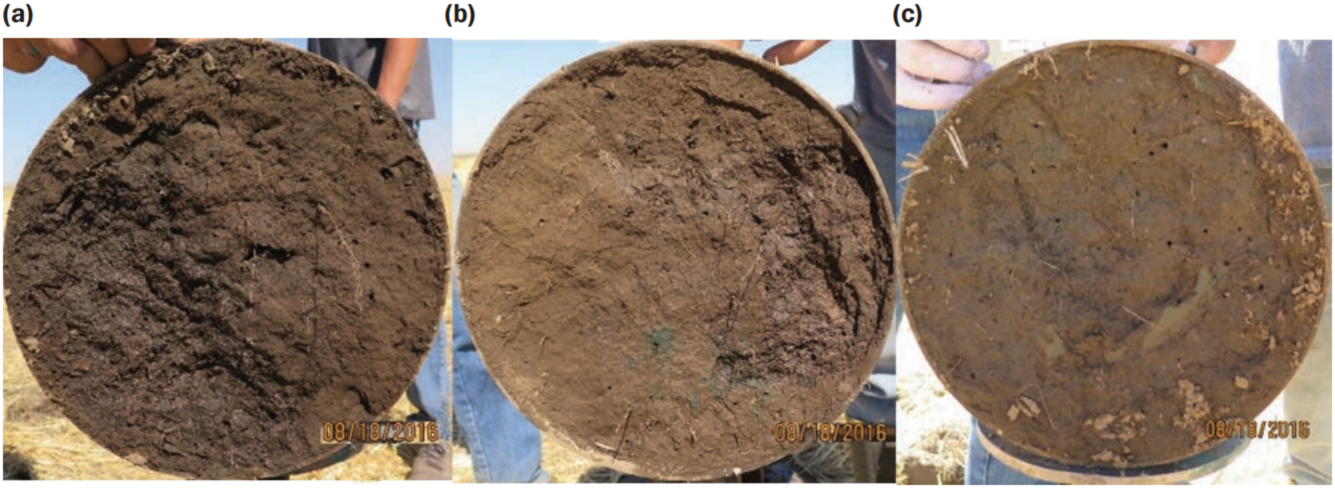
second seeding with double disk openers disturbed the soil surface, and tillage generally reduces aggregation and surface residue, leading to lower infiltration rates (Williams et al. 2018; Williams and Robertson 2016; Williams and Wuest 2011; Williams et al. 2014). The canola treatment infiltration rates, being relatively high in 2016 and substantially lower in 2017, indicate that reseeding probably did not have a substantial effect on infiltration. In 2018, infiltration rates for the canola treatment were relatively low compared to the 2016 measurements indicating that the successful fall canola seeding and more vigorous spring growth typical of a fall-seeded crop also did not enhance infiltration.

Similar to research reviewed by Cresswell and Kirkegaard (1995), no evidence was observed to support reports of oilseed roots creating biopores that influenced percolation. Neither dye-stained root channels nor large roots from either oilseed treatment were found during incremental removal of soil from within the cylinders. The lack of obvious biopores at sizes large enough to create substantial preferential hydrologic pathways is congruent with previous research on wheat–fallow and wheat–pea (*Pisum sativum*) rotations where the great majority of biopores were very small, and large pores were infrequent or concentrated near the surface (Wuest 2001a, 2001b).

Figure 4

Typical example of bottom ends of cylinders driven to 25 cm depth. (a) Canola treatment: saturated soil evident in the lower left third of core and no visible signs of large canola-sized root pores or dye concentration indicating biopore flow through the soil column. (b) Mustard treatment: signs of saturated flow on right side of ring with dye penetration in lower mid-area of ring, some of it near a pore. (c) Winter wheat treatment: no sign of saturated flow through ring, multiple macropores of indeterminant origin and little if any obvious flow through of dye.

(a) (b) (c)



Neither canola nor mustard provided a substantial boost to the productivity of subsequent spring wheat crops. It was thought that improvement in water infiltration or biological benefits of crop rotation might improve yield of the following crop. Although ponded infiltration rates after canola were substantially greater than after winter wheat, under natural rainfall conditions, the difference did not result in greater spring wheat yields. Our results were consistent with a six-year study conducted nearby where soil water recharge was not significantly different following harvest of winter canola versus winter wheat (Schillinger and Paulitz 2018). They actually measured a significant drop in spring wheat production following winter canola, which they could not specifically attribute to the previous winter canola crop.

There are several important agronomic factors other than water infiltration that need to be evaluated before recommending replacement of winter wheat–summer fallow with an intensified oilseed rotation. One is the amount of crop residue the brassica will leave behind. Enough durable residue needs to be produced to protect the soil from both wind and water erosion through winter, summer fallow, and the second winter during winter wheat crop establishment. In our study the oilseed crops (canola and mustard) were followed by a spring wheat crop, which produces 1.57 to 6.05 Mg ha^{-1} (0.7 to 2.7 tn ac^{-1}) crop residue (Papendick and Moldenhauer 1995), thus producing sufficient residue in productive years. However, Sharratt and Schillinger (2016) reported oilseed residue (standing and flat) values of 0.44 Mg ha^{-1} (0.2 tn ac^{-1}), which alone would not produce the 4 Mg ha^{-1} (1.8 tn ac^{-1}) needed to maximize soil protection and water infiltration if grown in a simple rotation of winter wheat–fallow–oilseed–fallow, and only during productive spring wheat years if grown in a winter wheat–oilseed–spring wheat–fallow rotation (Allmaras et al. 1980; Sharratt and Schillinger 2016).

The ability to measure differences in infiltration after one oilseed crop cycle suggests that oilseeds can have relatively quick effects on the soil, but further studies are necessary to determine whether the differences build or persist in complete rotations.

Summary and Conclusions

Ponded water infiltration rates were greater a year after canola and mustard crops compared to winter wheat. In the three years of measurement, there was no rotation benefit to yield on this level field where runoff is not expected. The difference in infiltration rates was not attributable to root channels.

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